

### Lavochkin La-5 FN

Most Soviet combat aircraft of the early war years were built entirely of wood, as light aircraft alloys were scarce. The La-5, with this weight penalty, was not quite a match for the speed of the Messerschmitt Bf 109G, but it was still extremely maneuverable. After the German invasion of the USSR in 1941, the Russians put enormous effort into expanding aircraft production. It was obvious that

not only sheer numbers, but also improved designs, were required to combat the skillful and well-equipped Luftwaffe. Yakovlev fighters were initially the most successful, while the LA-5, by contrast, was too slow. In 1942, The FN variant was reengined, transforming its performance and making it able to outperform even the Focke-Wulf Fw 190.

Wooden fuselage

|                                                                     |                                |
|---------------------------------------------------------------------|--------------------------------|
| <b>ENGINE</b> 1,700hp Shvetsov M-82FN air-cooled 14-cylinder radial |                                |
| <b>WINGSPAN</b> 32ft 2in (9.8m)                                     | <b>LENGTH</b> 27ft 11in (8.5m) |
| <b>TOP SPEED</b> 403mph (650kph)                                    | <b>CREW</b> 1                  |
| <b>ARMAMENT</b> 2 x 20mm ShVAK cannon above engine                  |                                |

### Lockheed P-38J Lightning

The Lockheed Lightning, designed for long-range missions, was the first fighter aircraft equipped with turbochargers and a tricycle undercarriage. It was complex and expensive to produce, and initially most went to the Pacific theater where long flights over water made the twin-engine arrangement desirable. Their most famous action there was in 1943, when 16 aircraft, operating 550 miles (885km) from their base, shot down the Japanese Commander-in-Chief, Admiral Yamamoto. Many Lightnings reached Europe in 1944 and were used by the tactical air forces for ground attack, and photographic reconnaissance.

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| <b>ENGINE</b> 2 x 1,425hp Allison V-1710 V-12 cylinders with turbochargers                                                       |                                 |
| <b>WINGSPAN</b> 52ft (15.9m)                                                                                                     | <b>LENGTH</b> 37ft 10in (11.5m) |
| <b>TOP SPEED</b> 414mph (666kph)                                                                                                 | <b>CREW</b> 1                   |
| <b>ARMAMENT</b> 1 x 20mm Hispano cannon and 4 x .5-in Browning machine guns; 3,200lb (1,452kg) bombload or 10 rocket projectiles |                                 |

1,425hp Allison engine with turbocharger

Twin-boom fuselage  
Twin-fin stabilisers

USAF markings

### Macchi M.C.202 Folgore

The Macchi M.C.200 Saetta was the main Italian fighter in 1940. It was a good dogfighter and able to perform well against the Hurricane, but it was underpowered. This disadvantage became crucial when faced with improving Allied types, and from 1941 the aircraft was equipped with a German engine – at first imported, and later built under license – to become the M.C.202 Folgore (“Thunderbolt”). Still not quite up to Allied or German standards, the Folgore served in North Africa, Sicily, and Russia until the Italian surrender in 1943.



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|------------------------------------------------------------------------|---------------------------|
| <b>ENGINE</b> 1,200hp Daimler-Benz DB 601A liquid-cooled V-12 cylinder |                           |
| <b>WINGSPAN</b> 34ft 9in (10.6m)                                       | <b>LENGTH</b> 29ft (8.9m) |
| <b>TOP SPEED</b> 369mph (594kph)                                       | <b>CREW</b> 1             |
| <b>ARMAMENT</b> 2 x 12.7mm and 2 x 7.7mm Breda-SAFAT MGs               |                           |

### Messerschmitt Bf 110C-5

Designed as a heavy, long-range fighter to protect strategic bombers, the Bf 110 was first put to the test in the Battle of Britain. It proved to be no match even for the Hurricane, which could easily outmaneuver it, and had to be protected, in turn, by Bf 109s. In other theaters it was very effective as a ground-attack aircraft, as long as there was no fighter opposition. It came into its own as a night fighter – equipped with airborne radar – from 1942 onward, inflicting significant losses on RAF Bomber Command raids over Germany.

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|-----------------------------------------------------------------------------------------|--------------------------------|
| <b>ENGINE</b> 2 x 1,100hp Daimler-Benz 601A-1 liquid-cooled V-12 cylinders              |                                |
| <b>WINGSPAN</b> 53ft 5in (16.2m)                                                        | <b>LENGTH</b> 39ft 9in (12.1m) |
| <b>TOP SPEED</b> 336mph (541kph)                                                        | <b>CREW</b> 2                  |
| <b>ARMAMENT</b> 4 x 7.9mm MG17 machine guns, 1 x 7.9mm MG15 machine gun in rear cockpit |                                |

### Mikoyan MiG-3

One of the few Soviet fighters available at the time of the German invasion in 1941, the MiG-3 performed creditably against the Luftwaffe onslaught. However, its overall weight restricted the armament that could be carried, and its performance below high altitude was poor.

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|------------------------------------------------------------------|-------------------------------|
| <b>ENGINE</b> 1,350hp Mikulin AM-35A liquid-cooled V-12 cylinder |                               |
| <b>WINGSPAN</b> 33ft 9in (10.3m)                                 | <b>LENGTH</b> 26ft 9in (8.2m) |
| <b>TOP SPEED</b> 398mph (640kph)                                 | <b>CREW</b> 1                 |
| <b>ARMAMENT</b> 1 x 12.7mm and 2 x 7.62mm machine guns in nose   |                               |

